

India Leads the Ship Recycling Sector



For Prelims: United Nations Conference on Trade and Development , National Shipbuilding Mission , International Maritime Organization , Shipbuilding Financial Assistance Scheme

For Mains: India's Maritime Vision 2030 and Maritime Amrit Kaal Vision 2047, India's shipbuilding ecosystem and Blue Economy, India-South Korea strategic partnership in the maritime sector

Source:TH

Why in News?

The **United Nations Conference on Trade and Development (UNCTAD)** reported that **India emerged as the world's leading ship recycling nation in 2025** , achieving a key target under the **Maritime India Vision (MIV) 2030** well ahead of schedule.

- Concurrently, the state visit by **South Korean President Lee Jae Myung to India in April 2026** has paved the way for accelerating India's broader ambition to become a global shipbuilding powerhouse.

Summary

- India has emerged as the world's largest ship recycling nation and is leveraging this advantage through major policy reforms, financial incentives, and strategic partnerships to transform itself into a global shipbuilding hub under Maritime India Vision 2030 and Maritime Amrit Kaal Vision 2047.
- Schemes such as SBFAS, SbDS, the Maritime Development Fund, and collaboration with South Korea aim to overcome structural bottlenecks, promote indigenous manufacturing, generate employment, strengthen maritime security, and position India among the world's top five shipbuilding nations by 2047.

What is the Current Status of India's Ship Recycling Sector?

- **Global Leadership:** India increased its share of global ship recycling activity from 30.1% in 2024 to **35.4% in 2025** .
 - The country recycled 2.99 million gross tonnes (GT) of ships, marking a nearly 60% increase from the previous year.
- **Target Achieved Early:** India achieved a key target of MIV 2030. Under MIV 2030 and **Maritime Amrit Kaal Vision 2047**, India aims to rank among the **top 10 global shipbuilding nations by 2030 and the top five by 2047**.
- **Policy Enablers:**
 - **Hong Kong Convention (HKC) Compliance:** Following the enactment of the **Recycling of Ships Act, 2019** , India aligned its sector with the **Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships**.
 - The government provided Rs 53 crore to modernize yards, enabling 115 facilities to become internationally compliant.
 - **Ship-breaking Credit Note Scheme:** This innovative scheme provides ship owners a credit note equivalent to 40% of the scrap value of a recycled ship.
 - This credit can be utilized (up to 5% of the vessel's value) to build a new ship at an Indian shipyard, directly linking recycling with domestic shipbuilding.
 - **Shipbuilding Financial Assistance Scheme (SBFAS):** **SBFAS** provides graded financial assistance (15% to 25% of the vessel cost) to domestic yards to offset cost disadvantages against subsidized foreign competitors.
 - It also establishes a **National Shipbuilding Mission** and introduces a **Shipbreaking Credit Note** worth **40% of the scrap value** to promote ship recycling and support new ship construction.
 - The scheme is expected to support **Rs 96,000 crore** worth of shipbuilding projects over the next decade.
 - **Shipbuilding Development Scheme (SbDS):** Focuses on expanding India's shipbuilding capacity through **greenfield shipbuilding clusters** , modernization of existing shipyards, and the establishment of an **India Ship Technology Centre** at the **Indian Maritime University (Tamil Nadu)** .
 - It provides **100% capital support** for common infrastructure in greenfield clusters and **25% capital assistance** for brownfield expansion, along with a **Credit Risk Coverage Framework** to improve project financing.
 - The SBFAS and SbDS aim to raise India's commercial shipbuilding capacity to **4.5 million Gross Tonnage (GT) per annum by 2047** , while promoting indigenous technology, employment generation, maritime security, and global competitiveness.
 - Both schemes will remain operational until **31 st March 2036** , with an **in-principle extension up to 2047** .
 - **Maritime Development Fund (MDF):** A dedicated Rs 25,000 crore vehicle to provide long-term, low-cost capital, comprising:
 - **Maritime Investment Fund (MIF):** A Rs 20,000 crore equity-based fund (with 49% government participation) to catalyze private and institutional capital.
 - **Interest Incentivization Fund (IIF):** A Rs 5,000 crore grant-based fund intended to heavily subsidize borrowing costs for shipbuilders over 10-to-15-year term loans, mitigating the crippling 10-11% domestic interest rates.
 - **Sagarmala Finance Corporation Limited (SMFCL):** India's **first dedicated Non-Banking Financial Company (NBFC)** tailored exclusively to underwrite maritime and shipbuilding projects.
 - **Green Tug Transition Programme (GTTP) & Harit Nauka Guidelines:** By mandating the phased

adoption of alternative fuels (LNG, battery-electric, hydrogen, methanol) for inland vessels and port crafts, the government is ensuring domestic yards secure an early-mover advantage in green shipbuilding.

- **Alang Expansion:** The **Alang Ship Recycling Yard** in Gujarat, the world's largest, is undergoing expansion to almost double its capacity to about 9 million light displacement tons (LDT).
- Furthermore, India is seeking inclusion in the **European Union's** approved list of ship recycling yards.

DEVELOPMENT OF SHIPPING AND MARITIME SECTOR IN 11 YEARS			
PARAMETER	2014	2025	% GROWTH
Cruise Passengers	84,000	4.92 lakhs	485%
Inland water cargo	18 MMT	146 MMT	700%
Operational waterways	3	29	870%
Investment in National waterways	Rs. 183 Cr	1,700 Cr	800%
Number of Seafarers	1.08 lakhs	3.2 lakhs	200%
Annual financial performance of ports	Rs. 11,172 Cr	Rs. 24,202 Cr	120%
No of new legislations	1 in 10 years	11 in 11 Years	

Source: Ministry of Ports, Shipping and Waterways

How Can South Korea Accelerate India's Shipbuilding Ambitions?

- **Developing Mega Shipbuilding Infrastructure:** Under the **India-ROK Comprehensive Framework, VOYAGES (Shared Vision for Operation of Yard Assisted Growth with Efficiency and Scale)** , South Korea is supporting the development of **India's first mega greenfield shipyard at Thoothukudi** , with a planned capacity of **2.5 million GT** and the potential to create around **15,000 direct jobs** .
- **Technology Transfer:** South Korea's leading shipbuilders - **HD Hyundai, Samsung Heavy Industries (SHI), and Hanwha Ocean** - are transferring advanced expertise in ship design, automation, green propulsion, and construction of high-value vessels.
- **Building a Localized Supply Chain:** The **Korea Marine Equipment Association (KOMEA)** , representing **over 300 enterprises** , has established a presence in **Mumbai** to promote domestic production of marine equipment and develop a localized shipbuilding supply chain.
- **Defence Manufacturing Cooperation:** Collaboration in naval and amphibious vessel manufacturing will strengthen indigenous defence production under the **Make in India** initiative.
- **Skill Development and R&D:** Partnerships involving **Korea International Cooperation Agency (KOICA)** , **India's Ministry of Ports, Shipping and Waterways (MoPSW)** , **Indian Maritime University (IMU)** and the **Korea Maritime & Ocean University (KMOU)** will enhance workforce skills and promote research in naval architecture and marine engineering.

- **Global Competitiveness:** South Korean investment, technology, and industrial expertise will help India modernize its shipyards, integrate into global value chains, and achieve its target of becoming a **top 10 shipbuilding nation by 2030** and **top five by 2047** .

Click here to Read: [India-Republic of Korea Strategic Vision 2026-2030](#)

What are the Key Challenges in the Indian Shipbuilding Sector?

- **India's Shipbuilding Deficit:** India handles nearly 95% of its trade volume and 70% of its trade value via maritime transport. However, its actual shipbuilding capacity stands at a meager **0.072 million Gross Tonnage (GT)** , capturing less than 1% of global output (ranking 18th globally).
 - Over 90% of global trade moves by sea, yet the **USD 150 billion annual shipbuilding market** is almost entirely monopolized. The "**Big Three**" (**China, South Korea, and Japan**) collectively control **95% of all new shipbuilding orders** by tonnage.
- **High Cost of Capital and Financial Constraints:** Shipbuilding is a highly asset-heavy industry, typically requiring 25% to 35% of the total vessel cost as working capital.
 - Indian shipbuilders face steep domestic interest rates averaging **10-10.5%** , compared to heavily subsidized rates of 4-8% in competitor nations.
 - Furthermore, Indian yards **lack access to direct sovereign refund guarantees**, placing them at a severe disadvantage when bidding for international commercial contracts.
- **Fragmented Ecosystem and High Import Dependency:** Unlike South Korea's highly integrated Ulsan cluster, India lacks a co-located network of marine ancillary suppliers.
 - Consequently, Indian yards import **60-70% of critical marine components** , including specialized shipbuilding-grade steel, advanced navigation suites, and propulsion engines.
 - This reliance inflates costs, disrupts Just-In-Time (JIT) manufacturing, and exposes yards to global supply chain shocks.
- **Infrastructure and Capacity Limitations:** Many domestic shipyards operate with restricted physical infrastructure. The **lack of mega dry docks** (most Indian dry docks are under 310 meters in length) completely restricts the country's ability to construct or repair **Ultra Large Crude Carriers (ULCCs)** and mega container ships, effectively locking Indian builders out of the most lucrative commercial shipping segments.
- **Weak Domestic Demand and Underdeveloped MRO:** A robust domestic order book is critical to achieving economies of scale. However, Indian shipping companies often prefer purchasing cheaper, readily available **second-hand foreign vessels** over ordering new domestic ships.
 - Additionally, **India holds less than a 1% share in the global Maintenance, Repair, and Operations (MRO) market**, as Indian-flagged vessels frequently divert to Singapore, Dubai, or Colombo for quicker, higher-quality servicing.
- **Specialized Human Capital Deficits:** While India has a massive engineering workforce, there is a critical shortage of personnel trained specifically in cutting-edge **marine engineering, automated ship design, and green propulsion technologies** .
 - This skills gap prevents the rapid absorption of advanced technologies transferred through international partnerships.
- **Regulatory and Bureaucratic Hurdles:** Despite recent ease-of-doing-business initiatives, persistent regulatory inconsistencies, **complex land acquisition processes for greenfield projects, and**

unpredictable legal timelines continue to dampen foreign investor confidence and delay long-term project execution.

What Measures are Needed to Strengthen India's Shipbuilding Sector?

- **The Ulsan Model:** Collaborations aim to replicate South Korea's highly successful "Ulsan model," which relies on a holistic, cluster-led development approach integrating shipyards with necessary ancillary industries and training institutions.
- **Integrating Recycling and Building:** India must leverage the steel and materials recovered from its world-leading ship recycling industry to feed its domestic shipbuilding supply chains, fostering a truly circular maritime economy.
- **Conditional Partnerships:** Foreign collaborations must be tied to mandatory local value-addition, technology transfer, and domestic skill development to ensure India becomes a technology owner, not just an assembler.
- **Aggressive Supplier Localization:** Focus must immediately shift toward localizing ancillary industries within the new SPV clusters to reduce the import bill for critical components.
- **Green Technology Leapfrogging:** Rather than merely playing catch-up in conventional shipping, India must capture the emerging market for next-generation vessels. Driven by the **International Maritime Organization's (IMO)** decarbonization mandates, initiatives like the **Green Tug Transition Programme** are essential to position India as an early mover in hydrogen fuel-cell and methanol/ammonia dual-fuel propulsion.
- **Strict Enforcement of RoFR:** The **Right of First Refusal (RoFR)** for Indian-built and Indian-flagged vessels must be strictly mandated in all government and Public Sector Undertaking (PSU) cargo tenders (such as ONGC or Coal India) to ensure a guaranteed domestic order book.

Conclusion

Strengthening the shipbuilding sector, often termed the "mother of heavy engineering"—is not merely an economic exercise but a strategic imperative. It holds massive multiplier effects for job creation (generating up to 30 lakh jobs) and upstream industries like steel and electronics. By synergizing massive capital outlays, strict policy enforcement for domestic demand, and the localized manufacturing ethos of *Aatmanirbhar Bharat*, India is poised to transition from a vulnerable consumer of global maritime logistics to a sovereign, world-class producer.

Drishti Mains Question:

'The future of India's Blue Economy depends on strengthening its shipbuilding ecosystem.' Critically examine.

Frequently Asked Questions (FAQs)

1. What is the Shipbuilding Financial Assistance Scheme (SBFAS)?

SBFAS provides **15-25% financial assistance** for vessel construction, establishes a **National Shipbuilding Mission**, and introduces **Shipbreaking Credit Notes** to boost domestic shipbuilding.

2. What is the objective of the Shipbuilding Development Scheme (SbDS)?

SbDS aims to expand India's shipbuilding capacity through **greenfield shipbuilding clusters**, **modernization of existing shipyards**, the **India Ship Technology Centre**, and a **Credit Risk Coverage Framework**.

3. How does the Shipbreaking Credit Note Scheme promote the circular economy?

It provides ship owners a credit equivalent to **40% of the scrap value** of recycled ships, which can be used for building new ships at Indian shipyards, linking ship recycling with shipbuilding.

4. Why is South Korea an important partner in India's shipbuilding ambitions?

Through the **VOYAGES framework**, South Korea supports India with **technology transfer**, **mega shipyard development**, **localized supply chains**, **skill development**, and **defence manufacturing cooperation**.

5. What are the major challenges facing India's shipbuilding industry?

The sector faces **high financing costs**, **dependence on imported marine components**, **inadequate infrastructure**, **weak domestic demand**, **limited MRO capacity**, **skill shortages**, and **regulatory bottlenecks**, affecting global competitiveness.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/EZCzWoLfNFs](https://www.youtube.com/embed/EZCzWoLfNFs)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. What is the importance of developing Chabahar Port by India? (2017)

- (a) India's trade with African countries will enormously increase.
- (b) India's relations with oil-producing Arab countries will be strengthened.
- (c) India will not depend on Pakistan for access to Afghanistan and Central Asia.
- (d) Pakistan will facilitate and protect the installation of a gas pipeline between Iraq and India.

Ans: (c)

Mains

Q. "Investment in infrastructure is essential for more rapid and inclusive economic growth." Discuss in the light of India's experience. (2021)

India-Seychelles Relations



For Prelims: Seychelles , Exclusive Economic Zone , Small Island Developing States , Indian Ocean Region , Sea Lanes of Communication , Ocean Thermal Energy Conversion

For Mains: India-Seychelles Bilateral Relations, India's Maritime Strategy in the Indian Ocean Region (IOR)

Source: PIB

Why in News?

The Prime Minister of India undertook a State Visit to **Seychelles** as the Guest of Honour for its 50 th National Day (Golden Jubilee of Independence) celebrations. The visit coincided with the **50 th anniversary of India-Seychelles diplomatic relations**, during which both countries unveiled a commemorative logo marking five decades of bilateral ties.

Summary

- Prime Minister's State Visit to Seychelles marked 50 years of diplomatic relations, resulting in landmark agreements on digital payments, healthcare, defence, space, agriculture, and maritime cooperation, further advancing India's MAHASAGAR Vision in the Indian Ocean Region.
- While Seychelles remains a key strategic partner for India's maritime security, Blue Economy, and Indo-Pacific interests, addressing challenges such as China's growing influence, limited trade, maritime security constraints, and climate vulnerabilities will be crucial for deepening bilateral ties.

What are the Key Outcomes of the Prime Minister's Visit to Seychelles?

- **Deployment of UPI:** An MoU between **NPCI International Payments Ltd (NIPL)** and the **Central Bank of Seychelles** will facilitate the rollout of **India's Unified Payments Interface (UPI)** in Seychelles, strengthening the digital payment ecosystem and boosting financial inclusion.
- **Umbrella Line of Credit:** The two sides have concluded an umbrella line of credit agreement which will be extended in rupees and amounts to Rs. 1250 crore. It will be used to support priority development projects in Seychelles.
- **Janaushadhi Scheme:** An agreement between **HLL Lifecare Limited** and the Ministry of Health of Seychelles will facilitate the export of highly affordable, high-quality generic medicines and medical supplies from India, integrating Seychelles **into India's flagship healthcare accessibility model**.

- A framework MoU was signed outlining the preliminary preparations and bilateral cooperation for the development of a state-of-the-art **New Seychelles National Hospital** .
- **Extradition Treaty:** A pivotal legal framework established between the two republics to streamline the **extradition of fugitives** and strengthen bilateral cooperation in **combating transnational crimes** , including drug trafficking and financial fraud.
- **Maritime Mobility:** An MoU between the respective Ministries of Ports and Transport allows for the mutual **recognition of training and certification of Indian seafarers** , enabling them to serve on Seychelles-flagged vessels and boosting maritime employment.
- **Diplomatic Capacity Building:** An institutional linkage was established between India's **Sushma Swaraj Institute of Foreign Service (SSIFS)** and Seychelles' **Ministry of Foreign Affairs and the Diaspora (MoFAD)** for joint training programs.
- **Space Cooperation:** An MoU for the exploration and use of outer space for peaceful purposes, focusing heavily on **satellite applications, remote sensing, space technology** , and capacity building for disaster management and climate monitoring.
- **Agricultural Research Work Plan (2026-2031):** An MoU between the **Indian Council of Agricultural Research (ICAR)** and the Seychelles Agriculture Department to accelerate joint research, technical study, and food security initiatives over the next five years.
- **Defence and Maritime Cooperation:** India handed over a ' **Made in India**' **Fast Patrol Vessel (FPV) named PS LESPWAR**, manufactured by Goa Shipyard Limited, to the Seychelles Coast Guard.
 - This asset aims to significantly enhance Seychelles' maritime surveillance and **Exclusive Economic Zone (EEZ)** patrol capabilities.
 - India also gifted ambulances, utility vehicles, and laser radial boats to support the country's disaster response, security, and developmental requirements.
- **'Guardian of the Blue Horizon' Award:** The President of Seychelles conferred the highest Presidential Distinction, '**Guardian of the Blue Horizon**' , upon the Indian Prime Minister.
 - It recognises his leadership in climate action, promotion of the **Blue Economy** , sustainable management of ocean resources, and efforts to advance the developmental aspirations of **Small Island Developing States (SIDS)** .
 - The award explicitly acknowledges India's global environmental initiatives, including the **International Solar Alliance (ISA)** , **Mission LiFE (Lifestyle for Environment)** , and the **International Big Cat Alliance** .
- **Vision MAHASAGAR:** These agreements operationalize **India's Vision MAHASAGAR (Mutual and Holistic Advancement for Security and Growth Across Regions)**.
 - By exporting successful domestic models like **UPI** and the **Janaushadhi Scheme** , India is moving beyond traditional infrastructure diplomacy. This "**Digital and Health Diplomacy**" approach reinforces India's position as a reliable, non-coercive development partner in the vital **Indian Ocean Region (IOR)** against the backdrop of shifting geopolitical dynamics.

Why is Seychelles Strategically Important for India?

- **Strategic Location in the Indian Ocean:** Seychelles is located along key **Sea Lanes of Communication (SLOCs)** and near major maritime chokepoints such as the **Mozambique Channel, Bab-el-Mandeb, and the Cape of Good Hope** , making it vital for maritime surveillance and trade security.
- **Countering China's Influence:** A strong partnership with Seychelles helps India counter **China's**

"String of Pearls" strategy in the Western Indian Ocean, preventing a strategic vacuum in the region.

- **Strengthening Maritime Security:** Seychelles is integrated into India's **Coastal Surveillance Radar Network (CSRN)** , with data linked to the **Information Fusion Centre-Indian Ocean Region (IFC-IOR)** , enhancing maritime domain awareness and cooperation against **piracy, maritime terrorism, IUU fishing, and drug trafficking** .
- **Advancing India's Maritime Vision:** Seychelles is a key partner in implementing India's **SAGAR** and **MAHASAGAR** initiatives, promoting capacity building, Blue Economy cooperation, and sustainable management of its **1.3 million sq km EEZ** .
- **Gateway to Africa:** As a member of the **African Union (AU)** and the **Indian Ocean Commission (IOC)** , Seychelles serves as a strategic bridge for India's diplomatic, economic, and security engagement with **East Africa** .

India-Seychelles Bilateral Relations

- **Diplomatic Relations:** India established diplomatic ties with Seychelles in 1976 after its independence, with **INS Nilgiri** participating in the country's Independence Day celebrations (29th June 1976).
- **Defence and Maritime Security:** India installed and maintains six **Coastal Surveillance Radar Systems (CSRS)** to help Seychelles police its vast Exclusive Economic Zone (EEZ).
 - The armies conduct the biennial joint military exercise **LAMITYE** . Seychelles actively participates in India-led maritime initiatives, including the **MILAN** naval exercise.
- **Trade, Investment & Connectivity:** Bilateral trade is modest but steady (~USD 73 million in 2024-25) and heavily skewed in India's favor. It is constrained primarily by the lack of direct shipping lines.
 - India exports rice, medical instruments, pharmaceuticals, and vehicles.
- **Human Capital:** Over 1% of the Seychellois population are alumni of the **Indian Technical and Economic Cooperation (ITEC)** program.
- **Climate Change & Energy:** As a founding member of the **ISA** , Seychelles collaborates heavily on renewable energy.
- **Diaspora and Cultural Ties:** The Indian diaspora is a highly influential demographic, comprising roughly **5,000 Persons of Indian Origin (PIOs)** and **7,000 Non-Resident Indians (NRIs)** within a total population of 120,000.



What are the Challenges in India-Seychelles Ties?

- **Geopolitical Competition:** The **Assumption Island naval project** remains stalled due to sovereignty concerns in Seychelles, while India's strategic interests in the Indian Ocean compete with China's growing economic and maritime presence.
- **Maritime Security Constraints:** Seychelles lacks the capacity to effectively monitor its **1.4 million sq km EEZ**, making it vulnerable to **illegal fishing (IUU), piracy, and drug trafficking**, despite India's security assistance.
- **Limited Economic Engagement:** Bilateral trade remains modest due to the small Seychellois market and poor connectivity, while delays in implementing **Indian-funded infrastructure projects** can affect India's strategic credibility.
- **Limited Institutional Capacity:** Seychelles' small administrative and technical workforce slows the implementation of large Indian-funded projects, affecting timely delivery and creating opportunities for competing powers.
- **Supply Chain Vulnerabilities:** Global disruptions, particularly from the **West Asia crisis**, have exposed Seychelles' dependence on external supplies, increasing India's role in providing emergency food and construction materials.

What Measures Can Strengthen India-Seychelles Relations?

- **Enhance Maritime Domain Awareness (MDA):** Deepen real-time intelligence sharing through the **IFC-IOR** to combat Illegal, Unreported, and Unregulated (IUU) fishing, piracy, and drug trafficking.
- **Scientific Diplomacy:** India should link its **Deep Ocean Mission** and the **CSIR-National Institute of Oceanography (NIO)** with Seychelles for sustainable bioprospecting. Joint research into marine genomics.
 - Discovering novel compounds from deep-sea organisms for pharmaceuticals and bio-materials can

transform the '**Blue Economy**' from traditional fisheries into a high-value, IP-driven biotech **partnership** compliant with the **Nagoya Protocol**.

- **Climate Finance Innovation:** Instead of merely offering grants, India can provide the technical and geospatial expertise required to audit, verify, and securitize these ecosystems into **Sovereign Blue Carbon Credits** .
 - By linking this to India's evolving carbon market frameworks and the Article 6 mechanisms of the Paris Agreement, **India can help Seychelles generate independent, non-debt capital**.
- **Energy & Climate Mapping:** ISRO should provide dedicated telemetry to map Seychelles' potential for **Ocean Thermal Energy Conversion (OTEC)** generating power from the temperature difference between deep and shallow waters.
 - Furthermore, real-time algorithmic mapping of coral bleaching and marine heatwaves positions **India as an indispensable technological partner for the archipelago's environmental survival**.

Drishti Mains Question:

Discuss the strategic importance of Seychelles in advancing India's maritime interests in the Indian Ocean Region.

Frequently Asked Questions (FAQs)

1. What is the significance of the MAHASAGAR Vision in India-Seychelles relations?

The **MAHASAGAR (Mutual and Holistic Advancement for Security and Growth Across Regions)** Vision promotes maritime security, sustainable development, digital connectivity, and capacity building across the Indian Ocean Region.

2. Why is Seychelles strategically important for India?

Seychelles occupies a strategic position along major **Sea Lanes of Communication (SLOCs)** and supports India's maritime security, **Blue Economy** , and Indo-Pacific strategy while countering China's growing influence.

3. What were the major outcomes of the Prime Minister's visit to Seychelles?

Key outcomes included agreements on **UPI deployment, Janaushadhi Scheme, Extradition Treaty, New Seychelles National Hospital, space cooperation, agricultural research, and defence cooperation** , including the handover of the **PS LESPWAR** patrol vessel.

4. What are the major challenges in India-Seychelles relations?

Key challenges include **China's strategic presence** , limited bilateral trade, maritime security constraints, delays in project implementation, and climate vulnerabilities faced by Seychelles as a **Small Island Developing State (SIDS)** .

5. How can India further strengthen its partnership with Seychelles?

India can deepen cooperation through **maritime domain awareness, scientific and Blue Economy partnerships, climate finance, digital connectivity, renewable energy, and capacity building** to strengthen long-term strategic ties.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/d9XEiiazesg](https://www.youtube.com/embed/d9XEiiazesg)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Mains

Q. 'The long-sustained image of India as a leader of the oppressed and marginalised nations has disappeared on account of its new found role in the emerging global order.' Elaborate. **(2019)**

India's Kharif Preparedness Amid El Niño



Source: PIB

In response to a significantly delayed southwest monsoon and a **43% rainfall deficit**, the Union Ministry of Agriculture & Farmers Welfare has proactively rolled out a **multi-layered contingency framework** to safeguard the upcoming **Kharif season against potential El Niño disruptions**.

- **Vulnerability Mapping and Prioritization:** The government has identified **315 districts across 12 states** (including Madhya Pradesh, Maharashtra, Gujarat, Uttar Pradesh, and Rajasthan) as potentially vulnerable to weak monsoon conditions.
 - Out of these, **111 high-priority districts** have been isolated due to critical **irrigation coverage falling below 25%**.
 - 76 districts are classified as medium priority (25-50% irrigation coverage), while 128 districts are low priority due to relatively better irrigation facilities.
- **Contingency Planning:** The **Indian Council of Agricultural Research (ICAR)** has formulated **District Agriculture Contingency Plans**. These localised, district-specific blueprints are being strictly transitioned from administrative files into actionable, field-level documents.
- **Climate-Resilient Crop Diversification:** States are mandated to promote **short-duration, low-water, and drought-resistant crop varieties**. Special cultivation emphasis is being shifted toward **pulses, oilseeds, and Shree Anna (millets)** to distribute risk and maintain agricultural yield in rainfed zones.
- **Macro Water Harvesting Framework:** Repair and desiltation of community assets like check dams, farm ponds, and reservoirs are being integrated aggressively with **MGNREGA** (now **VB-GRAMG**) to simultaneously create local employment and boost water-table recharge.
- **Three-Tier Financial Safety Nets:** The financial resilience of affected farmers is being anchored through the convergence of three flagship schemes: **Pradhan Mantri Fasal Bima Yojana (PMFBY)** for swift crop-loss insurance, accelerated disbursement of **Kisan Credit Cards (KCC)** for liquid working capital, and direct cash support via **PM-KISAN**.
- **Real-Time Institutional Monitoring:** To prevent data silos, a dedicated '**El Niño Monitoring Cell**' and a '**Crop Weather Watch Group**' have been institutionalized in New Delhi to coordinate weekly secretary-level and ministerial reviews alongside **731 Krishi Vigyan Kendras (KVKs)** for grassroots scientific advisories.
- **Food Security and Macro Targets:** Despite the monsoon anomalies, the Centre has maintained a **Kharif foodgrain production target of 176 million tonnes**, backed by comfortable national buffer stocks of wheat and rice to insulate the domestic market from supply shocks.

Read more: [Managing the El Niño Threat](#)

Birth anniversary of Bankim Chandra Chattopadhyay



Source: PIB

The Prime Minister of India paid homage to the great Indian novelist, poet, and journalist **Bankim Chandra Chattopadhyay (1838-1894)** on the occasion of his birth anniversary on **27 th June** .

- **Early Life:** Born on **27 th June 1838** in the village of Naihati in the Bengal Presidency (present-day West Bengal).
 - One of the **first graduates** of the University of Calcutta and served as **Deputy Magistrate and Deputy Collector** from 1858.
- **The Bengal Renaissance:** He is widely regarded as the **Father of the Modern Bengali Novel** and a leading figure of the **Bengal Renaissance** .
- **Major Literary Contributions:**
 - **Anandamath (1882):** A landmark historical novel inspired by the **Sannyasi Rebellion** of the late 18th century.
 - It became a powerful source of nationalist inspiration, and its patriotic themes attracted restrictions under British rule.
 - **Vande Mataram:** Composed in the **1870s** in **Sanskritised Bengali** , it was first published in **Bangadarshan (1875)** and later incorporated into **Anandamath (1882)** .
 - **Vande Mataram** was sung at the 1896 session of the Congress by Rabindranath Tagore.
 - It became the rallying cry of the **Swadeshi Movement (1905)** , and the **first two stanzas** were adopted as the **National Song of India** by the **Constituent Assembly in 1950** .
 - In 2026, the Ministry of Home Affairs (MHA) has directed that **Vande Mataram must be played or sung before the National Anthem, Jana Gana Mana**, whenever both are part of an official event.
 - **Other Notable Works:** **Durgeshnandini (1865)** (first major modern Bengali romance novel), **Kapalkundala (1866)** , **Vishabriksha (1873)** , **Devi Chaudhurani (1884)** , and **Krishna Charitra** .
 - The first Indian English novel is widely considered to be **Rajmohan's Wife (1864)** by Bankim Chandra.
 - **Journalism:** Founded the monthly magazine **Bangadarshan (1872)** , which promoted **Bengali identity and nationalist literature** . It was later revived by **Rabindranath Tagore** .
- **Role in the Indian National Movement:** Through his writings, he promoted **cultural nationalism** , popularized the idea of **Bharat Mata** , and inspired **deshbhakti (patriotism)** as the highest duty.
- **Legacy:** He passed away on 8 th April 1894. For his immense contribution to literature and the nation's intellectual awakening, he is revered as **Sahitya Samrat (Emperor of Literature)**.



Read more: [150th Anniversary of India's National Song](#)

China's LineShine Becomes World's Fastest Supercomputer



Source: TOI

China's **LineShine** supercomputer, based in **China**, has topped the latest TOP500 ranking, replacing **El Capitan** of the **United States** as the world's fastest publicly ranked supercomputer. **Frontier** ranked third, and **Aurora** ranked fourth, both from the **United States**, while **JUPITER Booster** from **Germany** completed the top five.

- **TOP500 List:** The TOP500 list, published twice a year since 1993, ranks the world's most powerful supercomputers using the **High Performance LINPACK (HPL) benchmark**, which measures how quickly machines solve complex mathematical problems.
- **About LineShine:** LineShine is located at the National Supercomputing Center in Shenzhen and has achieved a performance of around **2.19 exaflops**, meaning it can perform over **two quintillion calculations per second**.
 - Its debut expands the global exascale club from four to five systems and marks the first time **Asia, North America and Europe** simultaneously host HPL exaflop-class supercomputers.
- **Technical Uniqueness:** LineShine runs entirely on general-purpose **Central Processing Units (CPUs)** and is the **first TOP500 system to cross two exaflops** using a CPU-only architecture, unlike many Artificial Intelligence (AI)-focused systems that rely heavily on **Graphics Processing Units (GPUs)**.
- **HPC-AI Convergence:** LineShine reflects the growing merger of **traditional High-Performance Computing (HPC) and Artificial Intelligence (AI)**, as future scientific research increasingly requires systems that can handle both numerical simulations and machine-learning workloads.
- **Applications:** Used for advanced tasks such as **AI-assisted weather forecasting, rainfall prediction** across East Asia, and atomic-level simulations of magnetic materials.
- **Significance:** The achievement is significant because LineShine has been developed largely with Chinese technology, despite U.S. export restrictions on advanced AI chips and semiconductor tools.

Read More: [National Supercomputing Mission](#)

Netra Airborne Early Warning and Control (AEW&C) System



India's indigenous Netra Airborne Early Warning and Control (AEW&C) system has received Final Operational Clearance (FOC), marking its readiness for full operational use by the **Indian Air Force (IAF)**.

- **About:** Netra is India's **1st indigenously developed airborne surveillance and battle-management system**, developed by the Centre for Airborne Systems (CABS) under the **Defence Research and Development Organisation (DRDO)**.
- **Global Capability:** India is the 5th country, after the United States, Russia, Israel and China, to develop an indigenous AEW&C capability, strengthening its defence technology base and self-reliance.
- **Aircraft Platform:** The system is integrated on the **Brazilian Embraer EMB-145I aircraft platform** and functions as an **"eye in the sky"** by detecting, tracking, identifying and monitoring airborne as well as maritime targets.
- **Key Capabilities:** Netra is equipped with an **Active Electronically Scanned Array (AESA) radar**, Identification **Friend or Foe (IFF) system**, mission computer, secure communication networks, Electronic Support Measures (ESM), Communication Support Measures (CSM) and self-protection systems.
- **Operational Importance:** AEW&C systems act as force multipliers in modern warfare by providing early warning, real-time battlefield awareness and airborne command-and-control support for fighter aircraft and air-defence assets.
- **Historical Background:** India's airborne situational awareness programme began in the early 1980s under **Project Guardian**, using a modified HS-748 Avro aircraft as the Airborne Surveillance Platform (ASP); the programme suffered a setback after the 1999 Arakkonam crash and was later revived in 2004.
- **Role in Operations:** Netra proved its operational value during the 2019 Balakot strikes and **Operation Sindoor in 2025**, where it enhanced surveillance, situational awareness and network-centric operations of the IAF.
- **Significance** : Since the system is developed indigenously, India has greater control over its software, source codes, interface documents and upgrade pathways, allowing confidential capability enhancement without dependence on foreign manufacturers.
- **Future Expansion:** The Cabinet Committee on Security (CCS) has approved six more **upgraded Netra Mk-1A systems**, while the future **Netra Mk-2 programme on Airbus A321 aircraft** is expected to further strengthen India's airborne surveillance, network-centric warfare and defence self-reliance.

Read more: [India's Defense Modernization: Challenges and Opportunities](#)